

Introduction to Artificial Intelligence

LearnHub SonaMilkii — Original Study Notes

Learning Objectives

- Define artificial intelligence.
- Distinguish machine learning and deep learning.
- Identify common AI applications and responsible-use concerns.

1. What Is AI?

Artificial intelligence (AI) refers to computer systems designed to perform tasks that normally require aspects of human intelligence, such as recognizing patterns, understanding language, making predictions, or generating content.

2. Machine Learning

Machine learning is a branch of AI in which systems learn patterns from data to make predictions or decisions rather than relying only on explicitly written rules.

3. Deep Learning

Deep learning uses multi-layer neural networks to learn complex patterns. It is widely used in image recognition, speech processing, language tasks, and other applications.

4. Applications

AI is used in recommendation systems, search, translation, fraud detection, medical research, education tools, robotics, customer support, and content generation.

5. Responsible AI

Important concerns include privacy, bias, security, misinformation, transparency, and appropriate human oversight. AI outputs should be checked when accuracy matters.

Summary

AI is a broad field. Machine learning is one major approach, and deep learning is a machine-learning approach based on multi-layer neural networks.

Practice Questions

- What are the main ideas covered in Introduction to Artificial Intelligence?
- Which concept was most useful to you, and why?
- Give one practical example of applying what you learned.

Note: This document is an original educational study guide created for LearnHub SonaMilkii.